

I. Statement of Requirements

I.I. Mining Process

Sam & Ches is an online thrift store, primarily selling procured clothes & dresses locally and overseas via livestreams on the Facebook platform. Through this medium, sellers interact with their customers live from around the country in order to market their clothing and lock-in transactions. This process, in live-selling terms, is called *mining* which can be visualized below in *Figure 1*.

1. associates a code
with an item

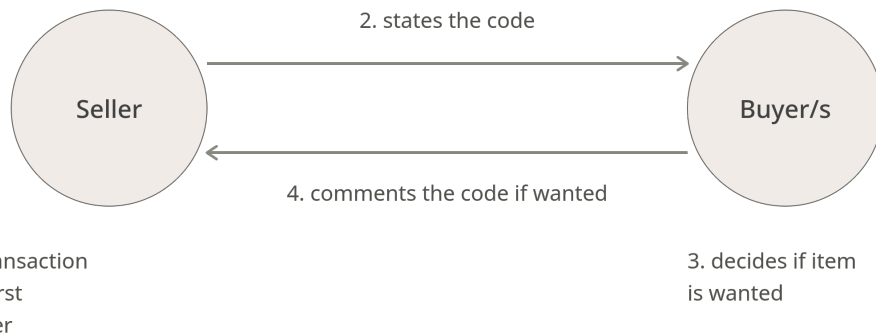


Figure 1.

Interpreting this figure, it can be seen that *mining* is a step-by-step process of locking a transaction with a customer, enumerating each of these are as follows:

- 1) *Associates a code with an item*
 - In this step, the seller starts with the assignment of a “code” to an item. This *code* will now be unique to this item until it has been locked for transaction. See step X to see how a transaction is locked.
- 2) *States the code*
 - The code to an item is then verbally transmitted to the customers through the livestream, acting as a feedback for the customers that the associated item is now up for sale.
- 3) *Decides if item is wanted*
 - The customer then decides if the item of the associated code is something they’re willing to buy.
- 4) *Comments the code if wanted*

- If the customer/s have finalized their decision in wanting to buy an item, the code is then re-stated by the customer/s via the livestream's comment section to let the seller or their staff know which customers are willing to make a transaction (*Note: customers may typically cancel at anytime before payment is made*)

5) *Locks transaction with the first commenter*

- The seller then reserves the item of the associated code with the customer that was able to first comment the code in the livestream.

In the backstage of doing these, it is typically tedious for livesellers to accomplish *Step 5* since, internally, there is much more going on. For *Sam & Ches*, they accomplish this step doing 3 more things:

- 1) Assigning a code and price to a piece of clothing via masking tape while live.
- 2) Manually monitoring the comments section to determine the “winner” of an item.
- 3) Encoding the customer and order details via excel.

To make matters harder, these steps are typically done under time pressure in order to maintain the pace of the livestream. Hence, the calling for a system to automate, at some capacity, a few of these steps to make it easier for *Sam & Ches* and possibly even other livesellers in accomplishing *Step 5*.

I.II. Processing of Digital Payments

Another issue that *Sam & Ches* faces is the processing of digital payments post stream. For many livesellers based in the Philippines, a popular mode of payment is in the form of digital transactions with the use of apps like *GCash & PayMaya*. As such, a model for processing digital payments can be visualized below in *Figure 2*.

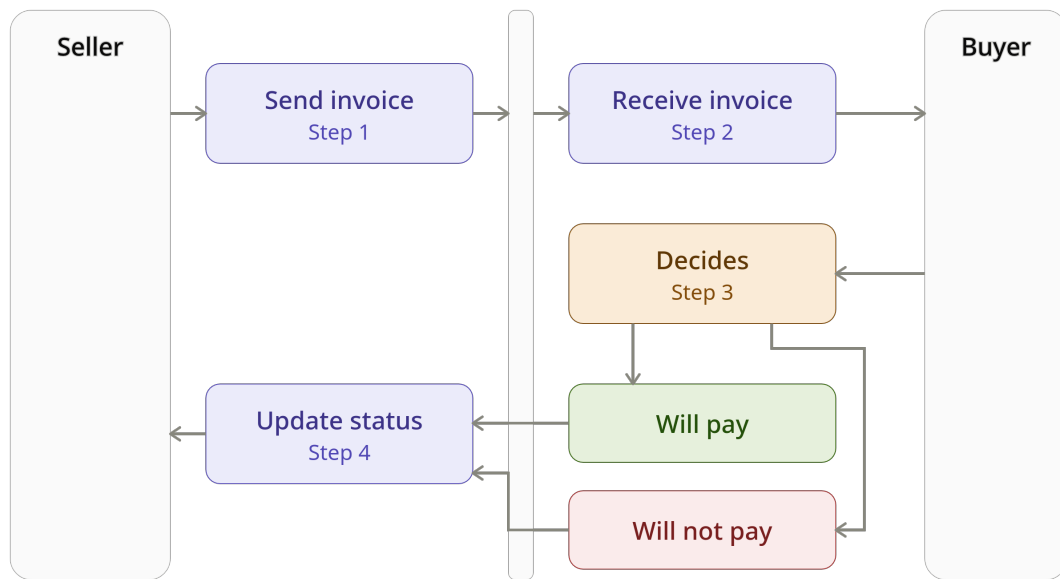


Figure 2.

Looking at the *figure* above the Seller and Buyer partake in another interaction during a transaction.

- 1) *Send invoice*
 - The seller sends an invoice detailing the ordered items a buyer has mined, their cost and modes of payment.
- 2) *Receives invoice & decides*
 - With the given invoice, the buyer decides whether they will continue buying it by either paying or not.
- 3) *Update status*
 - The seller then updates the status of the buyer's ordered items to either: paid or forfeited.

I.III. Order Status & Inventory Tracking

A final concern to be addressed is an improvement on order status and inventory tracking. Currently, *Sam & Ches* track the status of an order and their inventory via *spreadsheets*. This poses foreseeable issues in their efforts to expand due to the manual overhead required to encode and monitor business-related data. Some of these foreseeable issues include, but are not limited to:

- *Cost of Scalability*
 - The use of spreadsheets in expanding businesses get increasingly costly to maintain in relation with human resources. It gets more expensive to maintain a spreadsheet the bigger a business becomes.

- *Significantly Prone to Human Error*
 - As the customer base of a business grows, so is the probability for errors. Using spreadsheets will pose higher risk in managing inventory and orders.
- *Collaborative Unfriendly*
 - Managing inventory becomes increasingly difficult as more staff work on the spreadsheet. For an individual working on the spreadsheet, there may be many moving parts and changing states that make data difficult to track.

With that, below is *Table 1.* detailing the *requirements* and our proposed *solutions*.

Requirement	Solution
Automate capturing, recording and invoice generation of items mined by a customer.	Create an application that integrates with Facebook's features to: • Detect first comment matching an item code • Reserve the item • Generate an invoice for the item
Automate tracking & verification of payments for each item	Payment verification module utilizing third-party software to process digital payments. Possibly with a 1% fee. Ex. <i>Paymonggo</i>
Track item inventory & status	Integrate an inventory system in the application with multi-user access to managing <i>listings, claims and sales</i>

Table 1.

(Note that the requirements in the table are ordered by priority from top - bottom.)

II. Estimated Cost and Resources

Since *Sam & Ches*' proposed system is a moderately-scoped application (*comment-detection automation, payment verification, and inventory management*), the following estimates assume a small development team working over an academic-term timeline. Costs are based on typical entry-level developer rates in the Philippines and standard third-party service pricing.

II.I. Human Resources

Table 2. Roles and Responsibilities of each Roles

Role	Responsibility
Full-Stack Developer (<i>x2</i>)	Implements the Facebook comment-detection integration, invoice generation, and inventory module.
UI/UX Designer	Designs the seller dashboard and customer-facing order/payment views.
QA / Tester	Verifies code-matching accuracy, payment reconciliation, and inventory sync across multiple users.
Project Manager	Coordinates the timeline, documentation, and communication with <i>Sam & Ches</i> .

II.II. Development Cost Estimate

Table 3. Estimated Cost of the Project

Item	Estimated Cost
Development labor	₱0.00
Cloud hosting & database (<i>monthly</i>)	₱0.00 <i>Vercel Free Tier</i>
Payment gateway fees (<i>Paymongo</i>)	0.7–3.5% per transaction
Facebook API access	Free tier (<i>Meta Business verification required</i>)

III. Schedules and Milestones

The 14-week timeline in *Table 4* is mapped onto the First Semester calendar starting **September 7, 2026**, the Monday after the Project Plan submission, and ending **December 13, 2026**. Phases run sequentially and are time-boxed; if a phase runs out of time, scope is trimmed following the priority order of *Table 1* rather than extending the phase. Checkpoints with *Sam & Ches* are held at the end of every second week (*weekly when schedules allow*).

III.I. Phase Schedule

Table 5. Phase Schedule for Phase 1 Development

Phase	Weeks	Dates (2026)
Requirements & Design	W1 – W2	Sep 7 – Sep 20
Core Development (<i>mining automation, inventory</i>)	W3 – W8	Sep 21 – Nov 1
Payment Integration & Testing	W9 – W12	Nov 2 – Nov 29
Pilot Launch with <i>Sam & Ches</i>	W13 – W14	Nov 30 – Dec 13

III.II. Milestones

Each milestone marks a point where a phase, or a significant part of one, is considered **done**. The *exit criteria* define what must be true before a milestone is declared met.

Table 6. Project Milestones

ID	Milestone	Target	Exit Criteria
M1	Scope confirmed with client	W1 – Sep 13	Admin permission levels and feature priority from <i>Table 1</i> confirmed by <i>Sam & Ches</i> ; follow-ups from Meeting 1 closed.
M2	Requirements & design baselined	W2 – Sep 20	Requirements list, data model, architecture and tech stack agreed on; dashboard wireframes drafted; development environment set up.
M3	Mining loop working in development	W6 – Oct 18	A comment on a test livestream is detected, the first valid commenter is recorded, the item is reserved and an invoice is generated without manual steps.

ID	Milestone	Target	Exit Criteria
M4	Core Development complete	W8 – Nov 1	Inventory log, order status tracking, multi-user admin accounts and seller dashboard are functional; unit tests pass.
M5	Payment module integrated	W11 – Nov 22	An invoice can be paid through the payment gateway sandbox and the order status updates automatically; manual-confirmation fallback works.
M6	Testing complete	W12 – Nov 29	Integration and end-to-end test cases executed; no open high-severity defects; test report written.
M7	Client acceptance & handover	W14 – Dec 13	System deployed and connected to the client's Facebook page; staff onboarded; one live session run in parallel with the manual process; client signs off.

M1, M2, M4, M6 and *M7* are phase-gate milestones; the next phase does not begin until they are met. *M3* and *M5* are mid-phase checkpoints so that slippage in the two longest phases is caught early.

III.III. Gantt Chart

Phase / Milestone	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Requirements & Design														
Core Development														
Payment Integration & Testing														
Pilot Launch														
Client checkpoints														
Milestones M1 – M7														

Figure 3. Gantt Chart of Phase 1 (Sep 7 – Dec 13, 2026)

IV. Effort Breakdown

Effort is estimated in **person-hours** per major component of each phase. With 3 members committing at least 6 hours per week, team capacity is **18 person-hours per week**, or **252 person-hours** over 14 weeks. Estimates include learning time for unfamiliar technologies (*API integration, payment gateway's, etc.*)

IV.I. Effort per Phase

Table 7. Effort Estimate per Phase Component

Component	Estimated Effort (h)
Requirements & Design (W1 – W2)	
Requirements analysis & scope confirmation with client	10
System architecture, data model & tech stack decision	8
UI wireframes (<i>dashboard, inventory, invoice views</i>)	6
Development environment setup	6
<i>Subtotal: 30 h of 36 h capacity</i>	
Core Development (W3 – W8)	
Backend – Facebook API integration (<i>comment listener, first-comment detection</i>)	22
Backend – reservation, invoice generation, inventory & order status	24
Backend – authentication & multi-user admin accounts	8
Frontend – seller dashboard & inventory views	16
Unit testing & design documentation	10
<i>Subtotal: 80 h of 108 h capacity</i>	
Payment Integration & Testing (W9 – W12)	
Backend – payment gateway integration (<i>payment links, webhooks, manual fallback</i>)	20
Integration & end-to-end testing	12
Test report & defect fixing	12
Demo preparation	5
<i>Subtotal: 49 h of 72 h capacity</i>	
Pilot Launch with Sam & Ches (W13 – W14)	
Production deployment & connection to client's Facebook page	5
Staff onboarding & parallel run during a live session	9

Component	Estimated Effort (h)
Feedback collection & hotfixes	7
Handover documentation & acceptance meeting	5
<i>Subtotal: 26 h of 36 h capacity</i>	
Total: 185 h of 252 h capacity (73%)	

Effort is deliberately front-loaded into *Core Development*, which carries the technically riskiest work. The remaining 27% of capacity is left unallocated to absorb the learning curve, rework from client checkpoints, and reduced-capacity weeks around other course requirements and holidays.

V. Project Metric and Risks

To evaluate whether the system successfully addresses the requirements in *Table 1*, the following success metrics and risks are identified.

V.I. Success Metrics

- *Transaction Lock Time*
 - Average time from a customer commenting a code to the transaction being reserved, compared to the current manual process.
- *Encoding Error Rate*
 - Frequency of mismatched codes, duplicate reservations, or incorrect order details, compared to spreadsheet-based tracking.
- *Payment Reconciliation Accuracy*
 - Percentage of payments automatically matched to the correct order without manual follow-up.
- *Staff Time Saved*
 - Reduction in hours spent per livestream session on manual comment-monitoring and spreadsheet encoding.
- *Adoption Rate*
 - Proportion of *Sam & Ches'* staff actively using the system without reverting to manual methods after a defined period.

V.II. Risks

Table 8. Expected Risks in App Development

Risk	Likelihood	Impact	Mitigation
Facebook restricts or blocks automated comment-reading (<i>ToS/API limits</i>)	Medium	High	Use Meta's official Graph API where possible; avoid unauthorized scraping.
Payment gateway downtime or failed transactions	Low	High	Support fallback manual confirmation; monitor gateway status.
Staff resistance to adopting a new system over familiar spreadsheets	Medium	Medium	Provide training sessions and a gradual, parallel-run transition period.
Data privacy concerns (<i>customer names, contact info, payment records</i>)	Medium	High	Apply data encryption and comply with the Philippines' Data Privacy Act.
Scalability issues as livestream viewer/order volume grows	Low	Medium	Design with cloud-based, horizontally scalable infrastructure from the start.

VI. Sources.

Source used in Statement of Requirements

Used for: Improving figure for mining

<https://claude.ai/share/13f486b8-5c70-48e9-b280-ea9246848d50>

Source used in Statement of Requirements

Used for: Improving figure transaction

<https://claude.ai/share/952b6061-7cdc-4f67-bfca-eec479909ffb>

Source used in Statement of Requirements

Used for: Detailing inefficiency of spreadsheets in scaling businesses

<https://www.prodysey.solutions/blog/spreadsheets-vs-software>